

A sharp collapsed-fibre likelihood classification for a four-visible restricted Boltzmann machine

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<https://rsihouse.ai/openmath> ; <https://luma.com/yzp9abvr> ;

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Central source and adjudication archive: <https://github.com/alejandrozu/openmath-2026-judging>

Abstract

For pure four-parity data and a four-visible, one-hidden spin RBM, collapsed parameters are local likelihood maxima precisely below an explicit $\tanh^2$ threshold. At equality and beyond they are saddles in the full parameter space; every collapsed point remains globally suboptimal and non-strict.

The model and full-space theorem

For $s \in \{-1, 1\}^4$ and $0 < |\rho| \leq 1$, take data distribution $Q_\rho(s) = (1 + \rho s_1 s_2 s_3 s_4) / 16$. The one-hidden spin RBM has normalized probabilities proportional to $\exp(b \cdot s) \cosh(c + w \cdot s)$, with $b, w \in \mathbb{R}^4$ and $c \in \mathbb{R}$. Its expected log likelihood is ℓ_ρ . The collapsed fibre consists of $\theta_0 = (0, c_0, 0)$, which all represent the uniform visible distribution.

Every affine directional derivative at θ_0 is zero. The sharp theorem says θ_0 is a local maximum in the full nine-dimensional parameter space if and only if $\tanh^2(c_0) < (3 + 2|\rho|) / (3 + 6|\rho|)$. At equality or above, every full parameter neighbourhood contains both a strictly higher and a strictly lower likelihood point. Every θ_0 also has some finite RBM with strictly better likelihood and distinct equal-likelihood points arbitrarily nearby.

Why higher-order analysis is needed

The collapsed parametrization is singular. Ordinary stationarity and a quadratic Hessian test miss the parity interaction, whose first nonzero signal appears at higher order. The proof expands log cosh and decomposes the visible functions into Walsh/parity components. It controls nuisance visible biases and normalizing terms by quantitative Jensen and likelihood-domination estimates.

The quartic geometry produces the threshold. An independent ordinary fourth-order derivation agrees with the displayed expression, but that calculation alone does not prove a full neighbourhood maximum. The source's `StableRegion` and `LocalMaximum` lemmas supply uniform control over arbitrary small parameter perturbations, rather than just along one symmetric ray.

Outside the threshold, a quartic escape direction yields a higher point, while another direction gives a lower point. At the equality case, the proof moves along the likelihood-flat collapsed fibre to nearby parameters on the unstable side and transports saddle behaviour back to every neighbourhood. This is the subtle boundary step; it should be explained explicitly rather than inferred from a strictly signed quartic coefficient that vanishes there.

Non-strictness and global suboptimality

Changing the hidden bias while keeping $b=w=0$ leaves the visible distribution uniform, which creates a flat direction and makes collapsed local maxima non-strict. A suitable finite coupled RBM captures part of the nonzero parity signal and improves the likelihood. Thus a collapsed local maximum can be nonglobal even though it has nearby points of equal value.

The theorem concerns parameter-space likelihood only. It does not assert a distribution-space maximum, an attracting training basin, strict trapping, EM convergence or a lower bound on optimization time. It fixes exactly four visible spins, one hidden unit and pure parity data. Ordinary explorations for larger dimensions or general four-wise-uniform data are outside the selected formal endpoint.

Prior setting, formal closure and publication

Collapsed stationarity, singular mixture critical fibres and familiar RBM interaction formulas are prior mathematics. The Fukumizu-Akaho-Amari comparison identifies a degenerate situation where the second-derivative splitting matrix vanishes. The proposed delta is the exact higher-order threshold, including full-neighbourhood and boundary statements.

This publication pass independently replayed the exact 15-source frozen closure under Lean 4.33.1 and Mathlib 0df444a360eaa60ab8c11dca51a86af692955474, finishing on 5 October 2026 at 01:36:52 UTC. All sources and all three selected axiom prints passed. `sharp_parity_classification`, `collapsed_local_max_iff` and `nonstable_parameter_saddle` each use only `propext`, `Classical.choice` and `Quot.sound`, with no admitted, native-evaluation or unknown axioms. This verifies the displayed full-neighbourhood classification for four visible spins, one hidden unit and pure four-parity data, including its equality boundary; broader models and training claims remain outside its scope. The dated archive's separate internal 17-module post-recovery run is preserved as historical evidence rather than being relabeled as this independent run. Receipt: `Verification/sakana-opdp89-parity-sharp-fresh-build.json`.

This short manuscript records the construction and selected endpoints. The companion Original results anthology contains the extended ordinary proof exposition and its explicitly stated premises; the preserved Lean source remains the complete formal proof record.

Verification and reproducibility

Fresh execution: sakana-opdp89-parity-sharp: PASS; 15/15 custom modules compiled successfully; receipt Verification/sakana-opdp89-parity-sharp-fresh-build.json. Compiler pins, source hashes and endpoint axiom outputs are in Verification. Historical receipts and finite checks do not replace fresh replay.

ParityFour.sharp_parity_classification

```
theorem sharp_parity_classification (p c₀ : ℝ) (hp : 0 < |p|) (hp1 : |p| ≤ 1) :
  (∀ (b : Fin 4 → ℝ) (d : ℝ) (w : Fin 4 → ℝ),
    HasDerivAt (fun t => objective p (affineParameterRay c₀ b d w t)) 0 0) ∧
  (IsLocalMax (objective p) (nullParameter c₀) ↔
    Real.tanh c₀ ^ 2 < (3+2*|p|)/(3+6*|p|)) ∧
  ((3+2*|p|)/(3+6*|p|) ≤ Real.tanh c₀ ^ 2 → ParameterSaddle p (nullParameter c₀)) ∧
  (∃ θ : Parameters, objective p (nullParameter c₀) < objective p θ) ∧
  (∀ U : Set Parameters, U ∈ nhds (nullParameter c₀) →
    ∃ θ : Parameters, θ ∈ U ∧ θ ≠ nullParameter c₀ ∧
    objective p θ = objective p (nullParameter c₀))
```

Source: proofs/opdp89-parity-sharp/SharpClassification.lean:40.

Source SHA-256: 15ef1ad86c585c87e7ac4fabdc3a58702c915e69dbf6aa80d701258bc9efc602

ParityFour.collapsed_local_max_iff

```
theorem collapsed_local_max_iff (p c₀ : ℝ) (hp : 0 < |p|) (hp1 : |p| ≤ 1) :
  IsLocalMax (objective p) (nullParameter c₀) ↔
  Real.tanh c₀ ^ 2 < (3+2*|p|)/(3+6*|p|)
```

Source: proofs/opdp89-parity-sharp/SharpClassification.lean:22.

Source SHA-256: 15ef1ad86c585c87e7ac4fabdc3a58702c915e69dbf6aa80d701258bc9efc602

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